Homework 3

Graded

Student

Colin Skinner

Total Points

92.5 / 100 pts

Question 1

Problem 1

20 / 20 pts

✔ **– 0 pts** Correct

– 5 pts Incorrect differential equation

– 1 pt Algebra Error

– 5 pts Incorrect solution to differential equation

Question 2

Problem 2

20 / 20 pts

✔ **– 0 pts** Correct

– 1 pt Algebra Error

– 3 pts Physically impossible solution

– 3 pts Incomplete algebra

– 5 pts Formulation Error

– 10 pts did not finish the problem

Question 3

Problem 3

20 / 20 pts

✔ **– 0 pts** Correct

– 2 pts algebra error

– 5 pts Incorrect conservation of energy

– 10 pts Did not finish the problem

Question 4

Problem 4 — **12.5** / 20 pts

4.1 a — **10** / 10 pts

- ✔ **– 0 pts** Correct
- **– 2 pts** Sign error - momentums don't sum to 0
- **– 2 pts** Algebra mistake
- **– 5 pts** did not complete the problem
- **– 5 pts** Incorrection conservations

4.2 b — **2.5** / 10 pts

- **– 0 pts** Correct
- **– 5 pts** partially correct
- ✔ **– 7.5 pts** incorrect argument
- **– 10 pts** no attempt

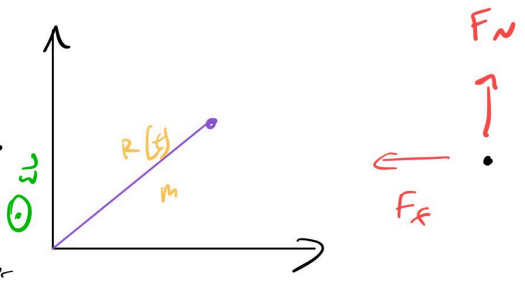
Question 5

Problem 5 — **20** / 20 pts

- ✔ **– 0 pts** Correct
- **– 5 pts** Does not state conservations
- **– 10 pts** No FBD/Force balance
- **– 5 pts** Does not attempt solution

Question assigned to the following page: 1

a constant
ube which
has a friction coefficient μ. Assuming initial conditions $r(0) = R$, $\dot{r}(0) = 0$, and no gravity, obtain the differential equation of motion and solve for $r(t)$, where r is the distance from the origin to the particle.



Energy not conserved due to friction.
Angular momentum irrelevant due to constraints.
No starting velocity so no linear momentum.

$r(t) = r\hat{r}$ $\quad v(t) = \dot{r}\hat{r} + r\dot{\hat{r}} = r\omega\hat{\theta} + \dot{r}\hat{r}$

$a(t) = \dot{r}\omega\hat{\theta} + r\dot{\omega}\hat{\theta} + r\omega\dot{\hat{\theta}} + \ddot{r}\hat{r} + \dot{r}\dot{\hat{r}} = \dot{r}\omega\hat{\theta} - r\omega^2\hat{r} + \ddot{r}\hat{r} + \dot{r}\omega\hat{\theta}$

$a(t) = (\ddot{r} - r\omega^2)\hat{r} + 2\dot{r}\omega\hat{\theta}$ only want

$\hat{\theta}$: $\sum F_\theta = ma_\theta \rightarrow F_N = ma_\theta \rightarrow F_N = ma_\theta = 2m\dot{r}\omega$

$\hat{r}$: $\sum F_r = ma_r \rightarrow F_f = \mu F_N = 2\mu m\dot{r}\omega = ma_r = m(\ddot{r} - r\omega^2)$

$2\mu\dot{r}\omega = \ddot{r} - r\omega^2 \rightarrow \ddot{r} + 2\mu\omega\dot{r} - \omega^2 r = 0$

2nd order ODE

$s_{1,2} = \frac{-2\mu\omega \pm \sqrt{4\mu^2\omega^2 + 4\omega^2}}{2} = \frac{-2\mu\omega \pm 2\omega\sqrt{\mu^2+1}}{2}$

$= -\mu\omega \pm \omega\sqrt{\mu^2+1}$

Assume solution

$r(t) = Ae^{s_1 t} + Be^{s_2 t}$

$r(0) = Ae^{s_1(0)} + Be^{s_2(0)} = R = A + B$

$\dot{r}(t) = As_1 e^{s_1 t} + Bs_2 e^{s_2 t} \rightarrow \dot{r}(0) = As_1 e^{s_1(0)} + Bs_2 e^{s_2(0)} = As_1 + Bs_2 = 0$

$\begin{bmatrix} 1 & 1 \\ s_1 & s_2 \end{bmatrix}\begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} R \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} A \\ B \end{bmatrix} = \frac{1}{s_2 - s_1}\begin{bmatrix} s_2 & -1 \\ -s_1 & 1 \end{bmatrix}\begin{bmatrix} R \\ 0 \end{bmatrix}$ $\quad A = \frac{Rs_2}{s_2 - s_1}$ $\quad B = \frac{-Rs_1}{s_2 - s_1}$

$r(t) = \frac{Rs_2}{s_2 - s_1}e^{s_1 t} - \frac{Rs_1}{s_2 - s_1}e^{s_2 t}$ where

$s_1 = -\mu\omega + \omega\sqrt{\mu^2+1}$

$s_2 = -\mu\omega - \omega\sqrt{\mu^2+1}$

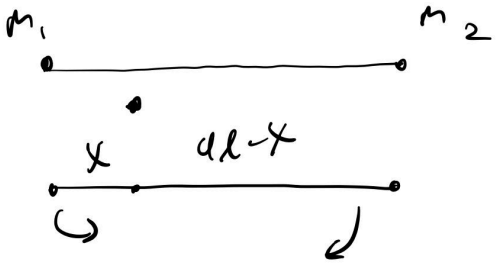
Question assigned to the following page: 2

string

paths

separated by a distance $4l$ on a smooth horizontal floor. Suddenly a point on the string at a distance l from particle m_1 strikes a fixed vertical nail of negligible diameter that projects from the floor. Assume that the particles proceed to whirl in opposite directions without colliding, and the string can slide freely on the nail. Find the maximum distance that m_1 can reach from the nail for the ensuing motion.

Hint: a particular quantity may be conserved for a single object, or a group of objects, or not at all. If you find yourself with more unknowns than equations, check if a certain quantity is conserved for multiple individual objects.



x: distance of strings shifting

Energy is conserved after collision

Angular momentum is conserved → Torque of radial force about nail is 0.

Initially

$H_1 = rmv = lmv_0$ $\quad H_2 = rmv = 3lmv_0$

$KE_i = 2\left(\frac{1}{2}mv_0^2\right) = mv_0^2$

After

$H_{1,f} = xmv_1 = lmv_0$ $\quad v_1 = \frac{lmv_0}{xm} = \frac{lv_0}{x}$

$H_{2,f} = (4l-x)mv_2 = 3lmv_0$ $\quad v_2 = \frac{3lmv_0}{(4l-x)m} = \frac{3lv_0}{4l-x}$

$KE_f = \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 = \frac{1}{2}m\left[\left(\frac{lv_0}{x}\right)^2 + \left(\frac{3lv_0}{4l-x}\right)^2\right] = mv_0^2$

$\frac{l^2}{x^2} + \left(\frac{3l}{4l-x}\right)^2 = 2 \rightarrow \frac{l^2v_0^2}{x^2} + \frac{9l^2v_0^2}{(4l-x)^2} = 2v_0^2$

$x = -0.78872, 1, 1.6532, 6.1355$ (between 1 and 4)

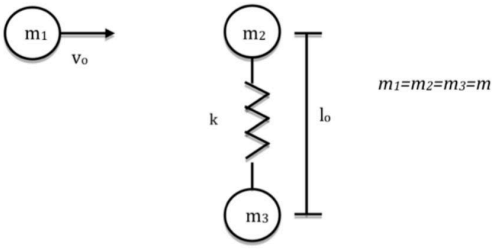
$x = 1.6532\,l$

Question assigned to the following page: 3

odules
catch
space debris so that the satellite can drag it down to a lower orbit. Normally the satellite is oriented vertically in space and the tether is unstretched. A piece of space debris (m_1) approaches the top end mass (m_2) at a relative speed of v_0 in a direction

perpendicular to the spring. It is caught in a net and sticks to the end mass m_2. In the ensuing motion, the tether stretches to a maximum length of $3l_0$. Solve for v_0 assuming that the masses (of the space debris and each of the two end modules) are equal and can be considered as particles. You can read more about this satellite concept here: http://www.space.com/12819-space-junk-cleanup-giant-net-tether.html

Hint: You can get one equation by considering conservation in the instants before and after collision.



Momentum conserved through collision.

- Linear because no ext. force
- Angular because no ext. torque

Energy conserved after collision with no ext. energy sources, but not through the collision due to inelastic.

Bottom mass at rest at collision moment.



$P_i = P_f$

$P_0 = mV_0 \quad P_{12} = 2mV_{12} \quad V_{12} = \frac{1}{2}V_0 \quad P_{com} = 3mV_{com} = mV_0 \quad V_{com} = \frac{1}{3}V_0$

At collision

$V_{12} = V_{com} + V_{12/com} \quad V_{12/com} = V_{12} - V_{com} = \frac{1}{2}V_0 - \frac{1}{3}V_0 = \frac{1}{6}V_0$

$V_{3/com} = V_3 - V_{com} = 0 - \frac{1}{3}V_0 = -\frac{1}{3}V_0$ but $|V_{3/com}| = \frac{1}{3}V_0$

$H_i = r \times p = \left(\frac{l_0}{3}\right)(2m)\left(\frac{1}{6}V_0\right) + \left(\frac{2l_0}{3}\right)(m)\left(\frac{1}{3}V_0\right) = \frac{1}{9}ml_0V_0 + \frac{2}{9}ml_0V_0 = \frac{1}{3}ml_0V_0$

$E_i = KE_{trans} + E_k + E_{rot}$

$E_i = \frac{1}{2}m_{12}V_{12}^2 = \frac{1}{2}(2m)\left(\frac{1}{2}V_0\right)^2 = \frac{1}{4}mV_0^2$

$KE_{trans} = \frac{1}{2}(3m)V_{com}^2 = \frac{1}{2}(3m)\left(\frac{1}{3}V_0\right)^2 = \frac{1}{6}mV_0^2$

At max stretch

$\dot{r}_2 = V_{2c} = \omega r_2 \quad \dot{r}_3 = V_{3c} = \omega r_3 \quad \omega = \frac{V_{2c}}{r_2} = \frac{V_{3c}}{r_3} \quad V_{3c}\left(\frac{2l_0}{3}\right) = V_{3c}\left(\frac{l_0}{3}\right) \quad V_{3c} = 2V_{2c}$

$H_f = r \times p = (l_0)(2m)(V_{2c}) + (2l_0)(m)(V_{3c}) = 2l_0m(V_{2c} + V_{3c}) = 6l_0mV_{2c}$

$H_i = H_f \quad \frac{1}{3}ml_0V_0 = 6l_0mV_{2c} \quad V_{2c} = \frac{1}{18}V_0 \rightarrow V_{3c} = \frac{1}{9}V_0$

$KE_{rot} = \frac{1}{2}(2m)V_{2c}^2 + \frac{1}{2}m(V_{3c})^2 = m\left(\frac{1}{18}V_0\right)^2 + \frac{1}{2}\left(\frac{1}{9}V_0\right)^2$

$= \left(\frac{1}{9^2}\right)\frac{1}{4}mV_0^2 + \left(\frac{1}{9^2}\right)\frac{1}{2}mV_0^2 = \frac{1}{9^2}\left(\frac{3}{4}\right)mV_0^2 = \frac{1}{108}mV_0^2$

Question assigned to the following page: 3

$2kl_0^2$

$E_i = kE_{trans} + E_k + kE_{rot}$

$\frac{1}{2}mv_0^2 = \frac{1}{6}mv_0^2 + 2kl_0^2 + \frac{1}{108}mv_0^2$

$27mv_0^2 = 18mv_0^2 + 216kl_0^2 + mv_0^2$

$8mv_0^2 = 216kl_0^2$

$v_0^2 = \frac{27kl_0^2}{m}$ $v_0 = l_0\sqrt{\frac{27k}{m}}$

$v_0 = 3l_0\sqrt{\frac{3k}{m}}$

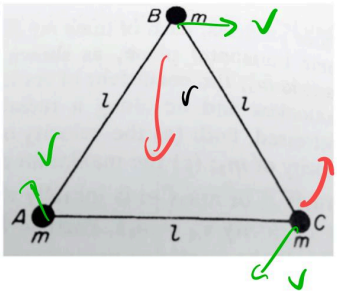
Question assigned to the following page: 4.1

connected by massless

, angular velocity ω_0 in planar motion about the motionless center of mass. Suddenly string AC breaks.

a. Solve for the particle speeds at the instant when the three particles first form a straight line.

b. Show that the particles A and C can never collide. Assume that strings AB and BC remain taut.



Energy conserved - no ext. forces

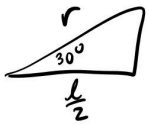
Angular momentum conserved

no ext forces.

+ direction

$\cos 30^\circ = \frac{\sqrt{3}}{2} = \frac{l}{2r}$

$r = \frac{l}{\sqrt{3}}$



so $v = \omega r = \frac{\omega l}{\sqrt{3}}$

$L_i = 0$ COM is not moving

$H_i = r \times p = 3\left[\frac{l}{\sqrt{3}} \cdot m \cdot \frac{\omega l}{\sqrt{3}}\right] = m\omega l^2$

$E_i = KE = 3\left[\frac{1}{2}m\left(\frac{\omega l}{\sqrt{3}}\right)^2\right] = \frac{3}{2}m\frac{\omega^2 l^2}{3} = \frac{1}{2}m\omega^2 l^2$

In a line the center of mass is in the middle.

$L_f = p = mv_A + mv_C + mv_B$ $L_i = L_f$

$\sqrt{3}m\omega l = mv_A + mv_C + mv_B$

Assumes that C goes in + direction and A does not

(1) $0 = v_A + v_B + v_C$

$H_f = r \times p = lmv_C - lmv_A$ $H_i = H_f$ $m\omega l^2 = lmv_C - lmv_A$

(2) $\omega l = v_C - v_A$

$E_f = \frac{1}{2}mv_A^2 + \frac{1}{2}mv_B^2 + \frac{1}{2}mv_C^2 = \frac{1}{2}m\left(v_A^2 + v_B^2 + v_C^2\right)$

$E_i = E_f$ $\frac{1}{2}m\omega^2 l^2 = \frac{1}{2}m\left(v_A^2 + v_B^2 + v_C^2\right)$ $\omega^2 l^2 = v_A^2 + v_B^2 + v_C^2$

(3)

(1) $v_B = -v_A - v_C$

Question assigned to the following page: 4.1

$$\omega^2 l^2 = V_A^2 + \left(-V_A - V_C\right)^2 + V_C^2$$

$$\omega^2 l^2 = V_A^2 + \left(V_A^2 + V_C^2 + 2V_A V_C\right) + V_C^2$$

$$\omega^2 l^2 = 2V_A^2 + 2V_C^2 + 2V_A V_C$$

(2) $-\omega l = V_C - V_A \rightarrow V_C = \omega l + V_A$

$$\omega^2 l^2 = 2V_A^2 + 2\left(\omega l + V_A\right)^2 + 2V_A(\omega l + V_A)$$

$$\omega^2 l^2 = 2V_A^2 + 2V_A^2 + 4\omega l V_A + 2\omega^2 l^2 + 2V_A \omega l + 2V_A^2$$

$$0 = 6V_A^2 + 6V_A \omega l + \omega^2 l^2$$

$$V_A = \frac{-6\omega l \pm \sqrt{36\omega^2 l^2 - 24\omega^2 l^2}}{12} = -\frac{1}{2}\omega l \pm \frac{\omega l \sqrt{12}}{6}$$

pick one

$$V_A = \frac{1}{2}\left(-1 + \frac{1}{\sqrt{3}}\right)\omega l$$

$$V_C = \omega l + V_A = \frac{1}{2}\left(-1 + \frac{1}{\sqrt{3}} + 2\right)\omega l$$

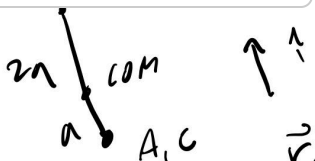
$$V_C = \frac{1}{2}\left(1 + \frac{1}{\sqrt{3}}\right)\omega l$$

$$V_B = -V_A - V_C = -\frac{1}{2}\left(-1 \pm \frac{1}{\sqrt{3}}\right)\omega l - \frac{1}{2}\left(1 \pm \frac{1}{\sqrt{3}}\right)\omega l$$

$$= \frac{1}{2}\left(\pm\frac{2}{\sqrt{3}}\right)\omega l$$

$$V_B = -\frac{1}{\sqrt{3}}\omega l$$

Question assigned to the following page: 4.2



$\vec{v}_A = -a\hat{\imath} = \vec{v}_C$

$\vec{v}_B = 2a\hat{\imath}$

At collision $\vec{v}_A = \vec{v}_C$

And H is conserved

$H_f = \vec{r}_A \times m\vec{v}_A + \vec{r}_B \times m\vec{v}_B + \vec{r}_C \times m\vec{v}_C$ v_A

$\vec{v}_B$ because $\vec{v}_A + \vec{v}_B + \vec{v}_C = 0$

$H_f = m\vec{r}_A \times (\vec{v}_A + \vec{v}_C) + \vec{r}_B \times m\vec{v}_B$

$= m(\vec{r}_A + \vec{r}_B) \times \vec{v}_B$ needs to be $v_B\hat{\imath}$ to keep strings taut

$m(r_A + r_B)\hat{\imath} \times v_B\hat{\imath} = 0$

but $H_f = 0 \neq H_i = mvl^2$

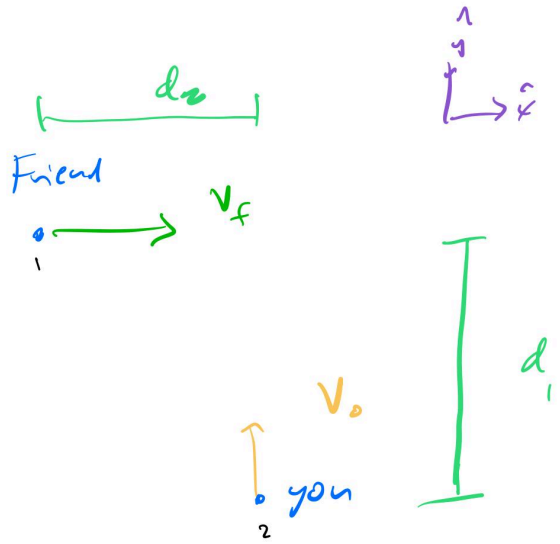
angular momentum conservation contradicts

Question assigned to the following page: 5

and began on the first homework, scription of what you are trying to solve. Again choose your reference frame and coordinate system and label it on your drawing. **Now determine what is and is not conserved and explain your reasoning (linear momentum, angular momentum and energy)**. Justify your answers clearly (yes or no, and for what objects in your system) using the methods discussed. Finally, "attempt" a solution to your problem. You may do this by using either a conservation law or through force balance. (Note! If you chose a difficult problem, you may not be able to solve your problem yet – hence the reason I asked you to choose a "simple" problem. In this case, attempt a solution and describe the reason why you cannot proceed beyond this point.)

A friend and I are biking past each other on campus. He is traveling at the speed v_f = 4 m/s and I am traveling at the speed 7 m/s. I am at a perpendicular distance of 14 m from his path of travel, and he is 18 meters from mine. At what point are we the closest, and at what angle should my friend wave to me clockwise from his direction of travel?

Adding a coefficient of friction of 0.15 for both bikes, and we have to find the speed of the bikes as well. Both of us weigh 150 kg. (hint: does one of the bikes come to a complete stop?)



- Energy is not conserved because there is an external source of energy
- Linear momentum is not conserved because there are external forces
- Angular momentum is conserved because there are no external forces

$\vec{r}_1 = r_1\hat{x}$ $\vec{v}_1 = \dot{r}_1\hat{x}$ $\vec{a}_1 = \ddot{r}_1\hat{x}$ $\Sigma F_{1x} = -\mu mg = ma_1 \rightarrow a_1 = \mu g$

$\vec{r}_2 = -r\hat{y}$ $\vec{v}_2 = -\dot{r}_2\hat{y}$ $\vec{a}_2 = -\ddot{r}_2\hat{y}$ $\Sigma F_{2x} = \mu mg = ma_2$ $a_2 = \mu g$

$x_1(t) = x_{1,0} + v_{1,0}t + \frac{1}{2}at^2$

$y_2(t) = y_{2,0} + v_{2,0}t + \frac{1}{2}at^2$

$\Delta x = \sqrt{x_1^2 + y_2^2}$ – s min when $x_1^2 + y_2^2$ is smallest, which is when $\frac{d}{dt}(x_1^2 + y_2^2)$ is the smallest

Question assigned to the following page: 5

$\frac{d}{dt}(x_1 + y_2) = 2x_1\frac{dx_1}{dt} + 2y_2\frac{dy_2}{dt}$

$0 = 2(x_{1,0} + v_{1,0}t + \frac{1}{2}at^2)(v_{1,0} + at) + 2(y_{2,0} + v_{2,0}t + \frac{1}{2}at^2)(v_{2,0} + at)$

$= x_{1,0}v_{1,0} + x_{1,0}at + v_{1,0}^2 t + v_{1,0}at^2 + \frac{1}{2}v_{1,0}at^2 + \frac{1}{2}a^2t^3$
$y_{2,0}v_{2,0} + y_{2,0}at + v_{2,0}^2 t + v_{2,0}at^2 + \frac{1}{2}v_{2,0}at^2 + \frac{1}{2}a^2t^3$

$= a^2t^3 + \frac{2}{2}(v_{1,0} + v_{2,0})at^2 + (x_{1,0}a + y_{2,0}a + v_{1,0}^2 + v_{2,0}^2)t + x_{1,0}v_{1,0} + y_{2,0}v_{2,0}$

$t = 2.7624\ s$ $\quad v_1 = -0.065\ \frac{m}{s}$ $\quad v_2 = 2.935\ \frac{m}{s}$

friend's bike comes to a stop

$KE_1 = E_f \rightarrow \frac{1}{2}mv_{1,0}^2 = Fd = \mu mg\,\Delta x_1$

$\Delta x_1 = \frac{v_{1,0}^2}{2\mu g} = \frac{(4\frac{m}{s})^2}{2(0.3)(9.8\frac{m}{s^2})} = 5.436\ m$ travelled

$x_{1,final} = x_{1,0} + \Delta x = -18m + 5.436m = -12.563m$

$y_{2,final} = 0$ when $y_{2,0} + v_{2,0}t + \frac{1}{2}at^2 = 0$ at $t_f = 2.859\ s$

when $x_{friend} = -12.5637\ m$ $\quad y_{me} = 0$

$\vec{v}_{friend} = 0$ $\quad \vec{v}_{me} = 2.935\hat{j}\ \frac{m}{s}$

with $\theta = 0°$